# CML — Phases (Chronic / Accelerated / Blast)

## 1. Three-phase pyramid

**WHAT YOU SEE**

central tiered pyramid: chronic at base, accelerated middle, blastic top exploding

**WHAT IT ENCODES**

Untreated CML progresses through three sequential phases — chronic → accelerated → blast crisis — driven by accumulating 'second-hit' mutations on top of BCR-ABL1. Each step up shortens survival and reduces TKI responsiveness. The whole point of TKI therapy is to freeze patients in chronic phase.

**BOARD TRAP**

Untreated CML inevitably advances to blast crisis within ~3-5 years; TKIs hold the overwhelming majority indefinitely in chronic phase with near-normal life expectancy.

## 2. Chronic phase — base

**WHAT YOU SEE**

blue chronic block, 90%, <5% blasts

**WHAT IT ENCODES**

Chronic phase = <10% blasts in blood and marrow (drawing shows <5%); ~90% of patients present here. Often asymptomatic (incidental leukocytosis) or with fatigue, splenomegaly, and early satiety. Highly responsive to TKIs.

**BOARD TRAP**

Diagnosis-stage CML is almost always chronic phase: massive leukocytosis with full myeloid spectrum, basophilia, low LAP, and splenomegaly — not a blast-dominant acute picture.

## 3. Chronic <5% blasts

**WHAT YOU SEE**

<5% BLASTS tag on the calm blue base block of the pyramid (a meditating figure sits there) — the stable, low-blast chronic foundation

**WHAT IT ENCODES**

Chronic-phase blast count is low (<10% in marrow/blood). Serial CBCs should stay stable; a rising blast or basophil count is the earliest sign of transformation.

**BOARD TRAP**

Rising blasts/basophils on serial counts signal transformation — recheck the marrow karyotype for clonal evolution and reassess TKI adherence/resistance.

## 4. Accelerated phase — middle

**WHAT YOU SEE**

rainbow accelerated band, 60–70% 8-yr

**WHAT IT ENCODES**

Accelerated phase is the intermediate, less-stable stage. WHO/ELN criteria: blasts 10-19%, peripheral basophils ≥20%, persistent thrombocytopenia (<100k) unrelated to therapy or uncontrolled thrombocytosis, increasing spleen/WBC unresponsive to therapy, or clonal evolution (new ACAs).

**BOARD TRAP**

Accelerated criteria to memorize: blasts 10-19%, basophils ≥20%, persistent thrombocytopenia <100k (or refractory thrombocytosis), and clonal evolution. Any one of these upstages from chronic phase.

## 5. 10–19% blasts (accelerated)

**WHAT YOU SEE**

10–19% BLASTS marker on the orange/rainbow accelerated band — the unstable middle window between chronic (<10%) and blast crisis (≥20%)

**WHAT IT ENCODES**

Blasts of 10-19% in blood or marrow defines accelerated phase. This is the window between chronic (<10%) and blast crisis (≥20%).

**BOARD TRAP**

≥20% blasts = blast crisis; 10-19% = accelerated; <10% = chronic. The 20% cutoff is the single most tested number in CML staging.

## 6. ≥20% basophils

**WHAT YOU SEE**

≥20% baso marker on accelerated band

**WHAT IT ENCODES**

Peripheral basophils ≥20% is a standalone accelerated-phase criterion. Basophilia is a CML hallmark throughout, but the ≥20% threshold specifically marks acceleration.

**BOARD TRAP**

Don't confuse the two basophil-related numbers: basophils ≥20% = accelerated phase; this is separate from the blast thresholds.

## 7. Clonal evolution

**WHAT YOU SEE**

CLONAL EVOL red tag among the accelerated-phase criteria boxes — new karyotype abnormalities that push the disease up the pyramid

**WHAT IT ENCODES**

Clonal evolution = emergence of new cytogenetic abnormalities in the Ph clone beyond the original Ph chromosome. It is an accelerated-phase criterion and predicts progression.

**BOARD TRAP**

High-risk additional abnormalities (+8, i(17q)/iso17q, double Ph, 3q26.2/EVI1) signal progression — detected only on full karyotype, not FISH/PCR.

## 8. Blast phase / crisis — top

**WHAT YOU SEE**

exploding blastic top, ≥30%

**WHAT IT ENCODES**

Blast crisis = ≥20% blasts in blood/marrow by WHO (the drawing's ≥30% reflects older ELN criteria). It behaves like an acute leukemia, responds poorly, and carries a dismal prognosis (median survival measured in months without transplant).

**BOARD TRAP**

WHO uses ≥20% blasts for blast phase; older/ELN criteria use ≥30%. Extramedullary (e.g. CNS, skin, bone) blast proliferation also counts as blast crisis regardless of marrow blast %.

### 9. Myeloid 60% blast crisis

**WHAT YOU SEE**

myeloid 60% segment

**WHAT IT ENCODES**

~60-70% of blast crises are of myeloid lineage and are treated with AML-type induction chemotherapy plus a TKI, ideally bridging to allogeneic HSCT. Responses are short.

**BOARD TRAP**

Myeloid blast crisis is treated like AML and has a WORSE response/prognosis than lymphoid blast crisis.

---

### 10. Lymphoid 25% blast crisis

**WHAT YOU SEE**

lymphoid 25% segment

**WHAT IT ENCODES**

~20-30% of blast crises are lymphoid (usually B-lineage) and are treated with ALL-type regimens (e.g. hyper-CVAD) plus a TKI, then HSCT. Better response (~CR 70%) and prognosis than myeloid.

**BOARD TRAP**

Lymphoid blast crisis responds BETTER to ALL-type therapy + TKI than myeloid blast crisis does — a classic 'which has the better prognosis' question.

---

### 11. HSCT cure

**WHAT YOU SEE**

HSCT CURE flag planted near the exploding blastic top — transplant reserved for the advanced apex, not the stable base

**WHAT IT ENCODES**

Allogeneic HSCT is the only proven curative modality but is reserved for TKI-resistant disease, T315I where ponatinib/asciminib are unavailable, and accelerated/blast phase. Cure rates fall sharply with advanced phase (CP 50-70% → BP ≤20%).

**BOARD TRAP**

HSCT is NOT first-line for chronic-phase CML — TKIs are. Transplant is for advanced/blast phase or TKI failure/resistance; transplant earlier (better outcomes in less-advanced disease).

---

### 12. Diagnosis demographics

**WHAT YOU SEE**

left wall: 15%, age 55–65, 9,000/yr, M:F 1.6:1

**WHAT IT ENCODES**

CML is ~15% of adult leukemias, median age 55-65 (rare in children), ~9,000 new US cases/yr, slight male predominance (M:F ~1.6:1). Ionizing radiation is the only established environmental risk.

**BOARD TRAP**

The Philadelphia chromosome t(9;22)/BCR-ABL1 is REQUIRED for the diagnosis — a 'CML' without demonstrable BCR-ABL1 is atypical CML or another MDS/MPN overlap, not classic CML.

---

### 13. TKI era survival

**WHAT YOU SEE**

85% TKI-era survival vs pre-TKI 3–6 yr

**WHAT IT ENCODES**

TKIs revolutionized prognosis: imatinib (IRIS trial, 2001) converted CML from a fatal disease into a chronic one. Chronic-phase patients now have ~85-90% long-term survival approaching that of the general population.

**BOARD TRAP**

Pre-TKI median survival was only ~3-6 years (busulfan/hydroxyurea/interferon era); imatinib changed the natural history — this before/after contrast is heavily tested.

---

### 14. Response milestones bar

**WHAT YOU SEE**

bottom survival bar: PCyR, CCyR, MMR, MR4, MR4.5

**WHAT IT ENCODES**

Response is monitored by BCR-ABL1 transcript milestones on the IS: ≤10% at 3 months, CCyR (≤1%) by 12 months, MMR/MR3 (≤0.1%) as the long-term goal, and deep responses MR4 (≤0.01%)/MR4.5 (≤0.0032%) enabling treatment-free remission attempts.

**BOARD TRAP**

Failure to hit milestones (e.g. >10% at 3-6 months, >1% at 12 months) should trigger: check ADHERENCE first, then drug interactions, then BCR-ABL1 kinase-domain mutation testing, then switch TKI.

---